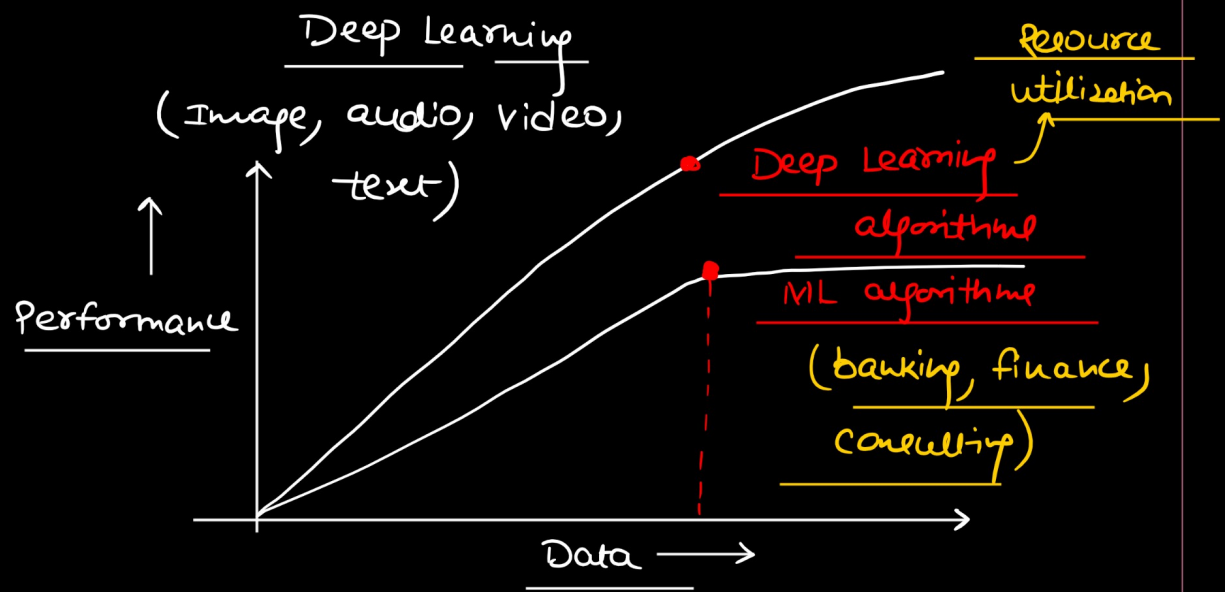




Neural Network

→ to learn the patterns

Input layer

Hidden layer

Output layer

Image →

Audio →

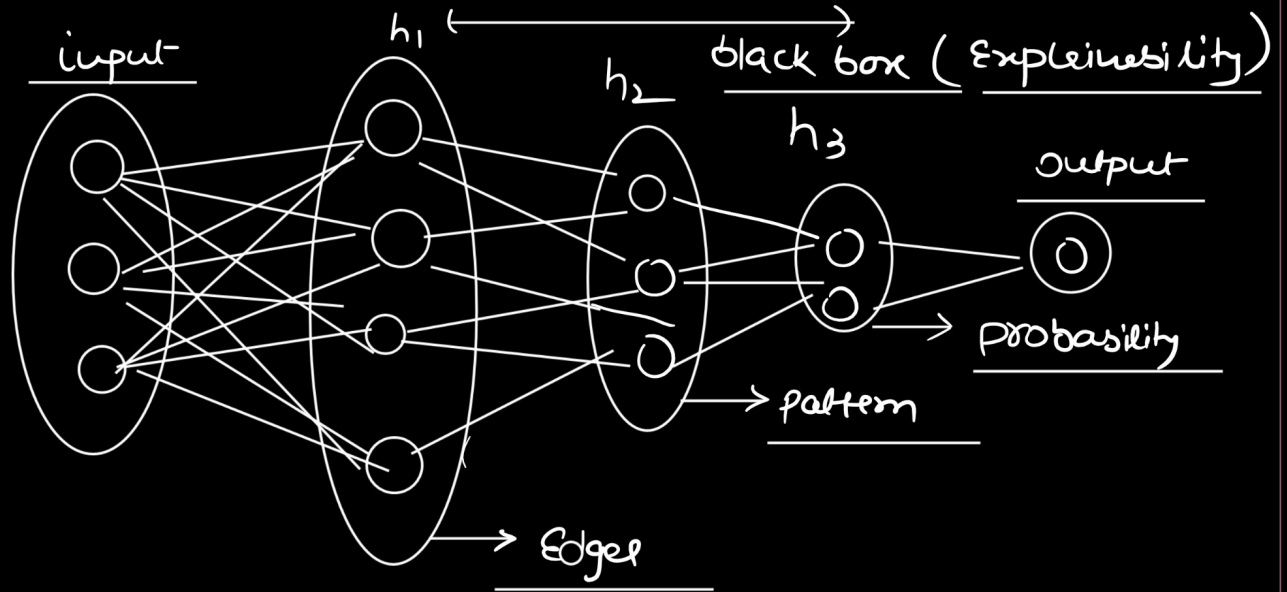
Text →

Video →

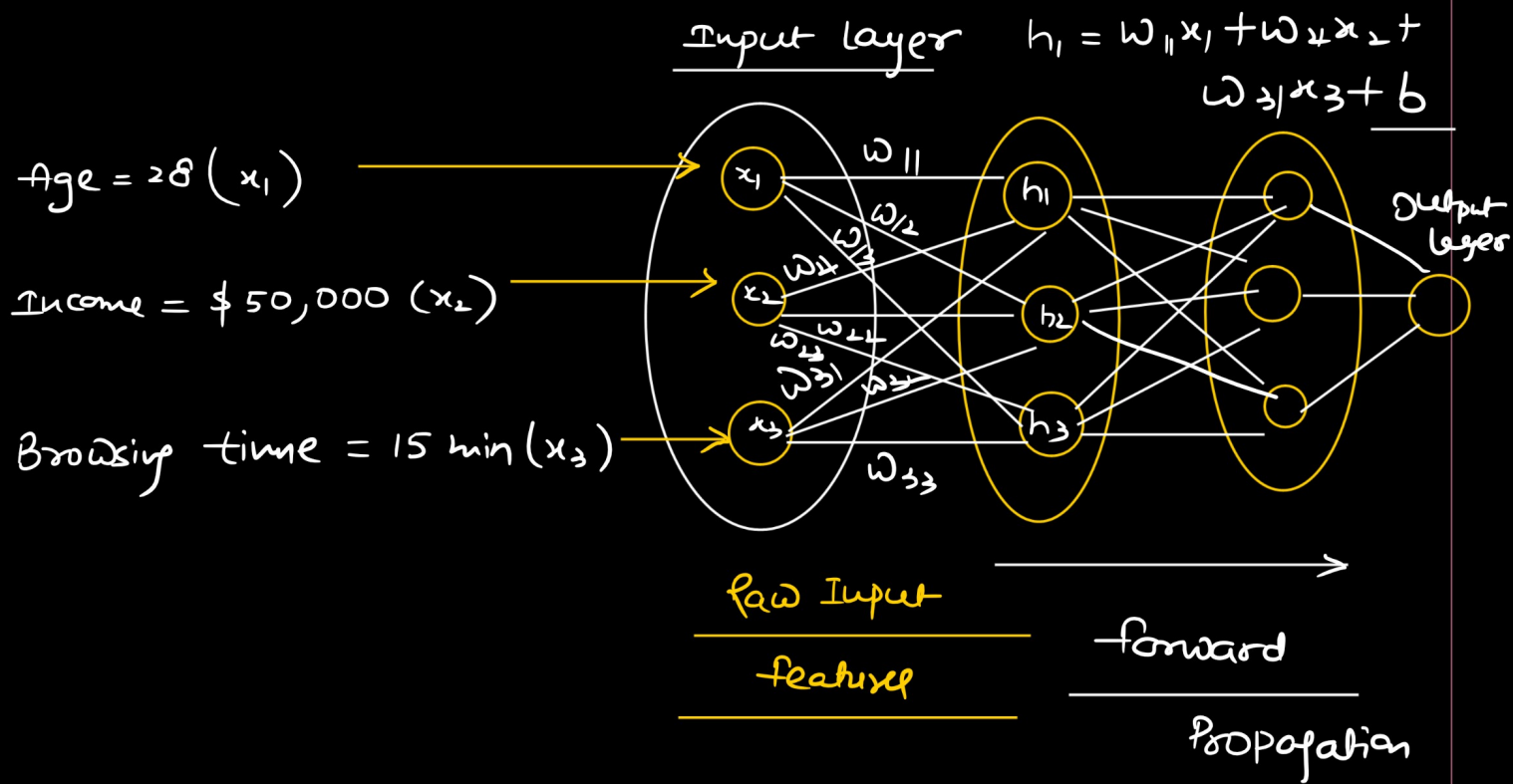
hidden layers

neurons inside hidden layers

Input layer



How the learning happens?



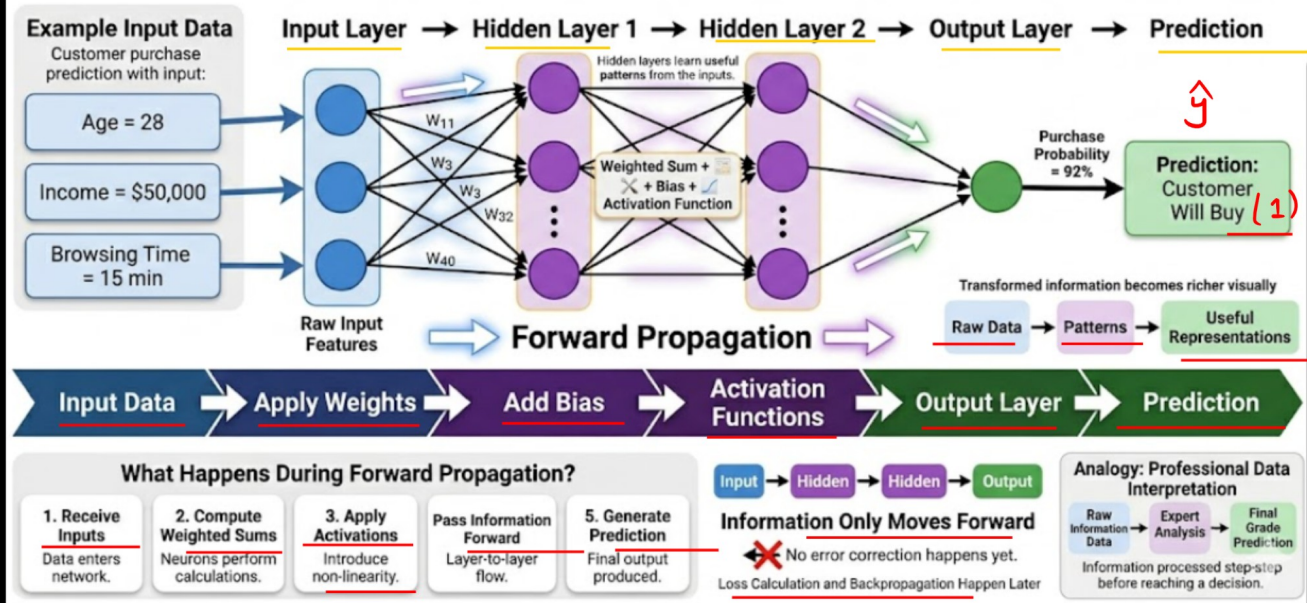
$$\Rightarrow \sum w_{ij}x_i + b_i \text{ (Linear Equation)}$$

To add non-linearity $\rightarrow$ ReLU, tanh, sigmoid

$\Rightarrow$ Activation function

Forward Propagation Intuition

How Information Flows Through a Neural Network



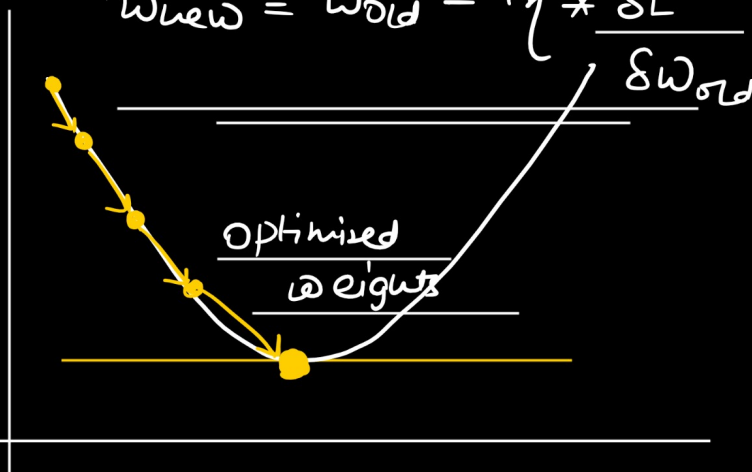
⇒ Random initialization of weights

Reduce the Overall Error $\sum_{i=1}^n (y_i - \hat{y}_i)$

$y \approx \hat{y}$ (Loss / Residual / Error)

Optimize → Gradient Descent

$$w_{new} = w_{old} - \eta * \frac{\partial L}{\partial w_{old}}$$



Optimizer

→ to

Optimize the

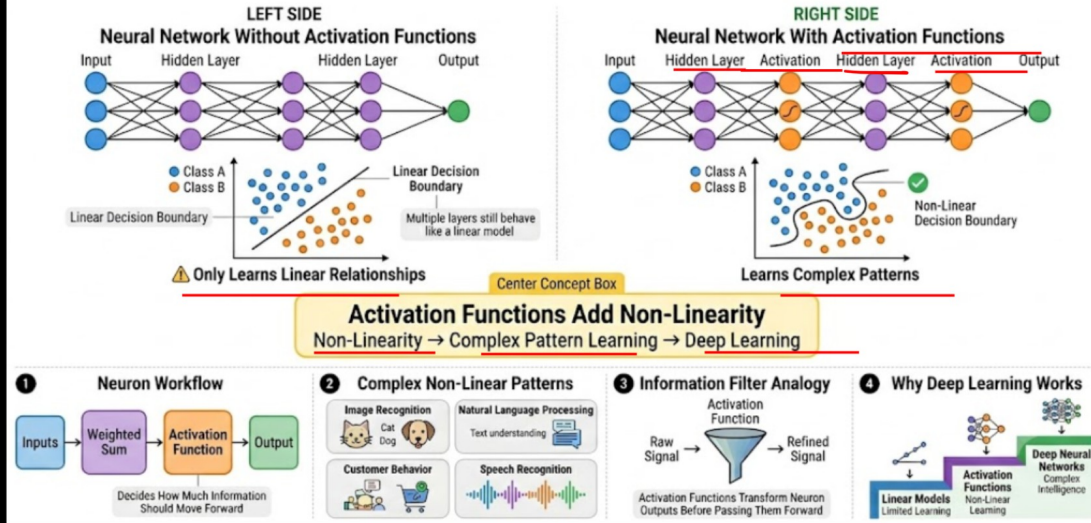
Weights in the

neural network

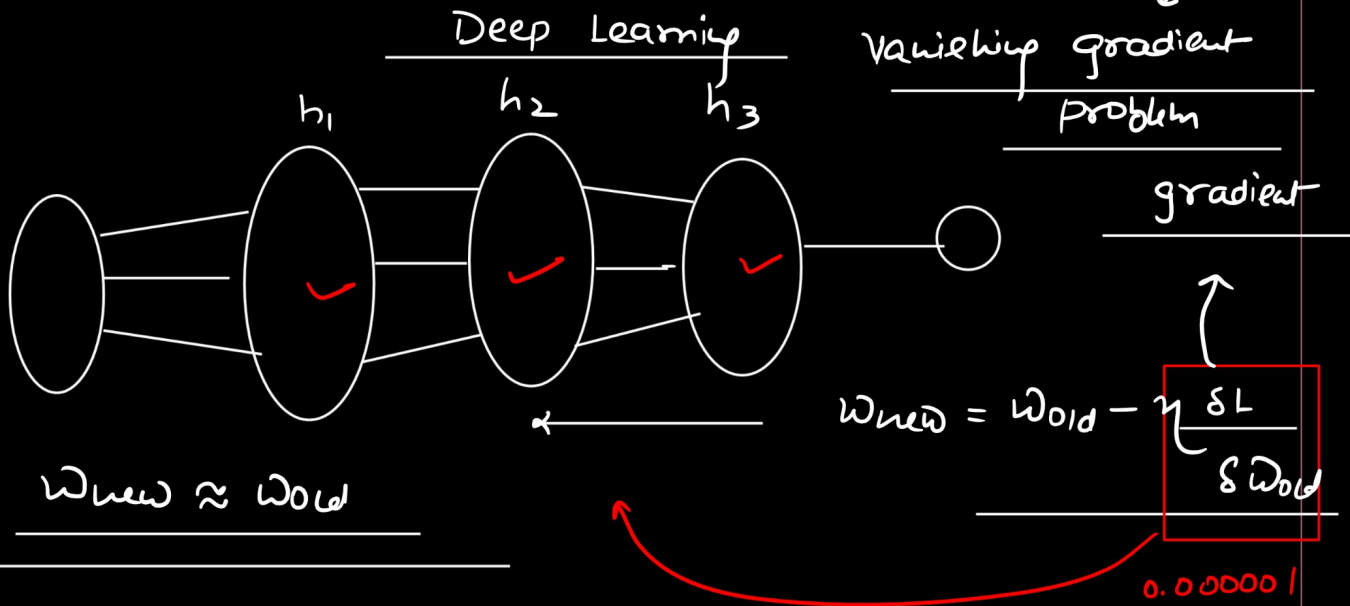
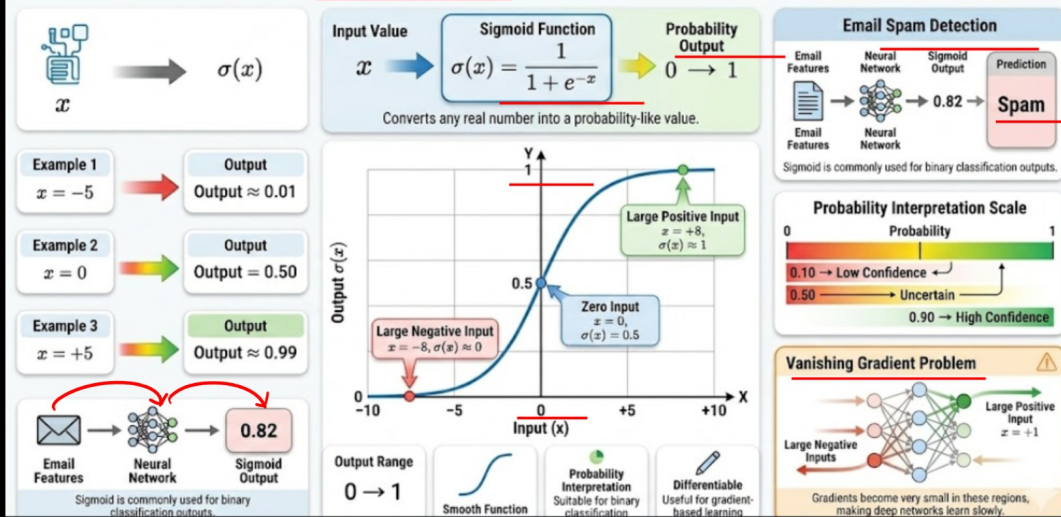
to get overall loss error

Why Activation Functions Are Needed

How Neural Networks Learn Complex Non-Linear Patterns



Sigmoid Activation Function



tanh, ReLU

hidden neural network

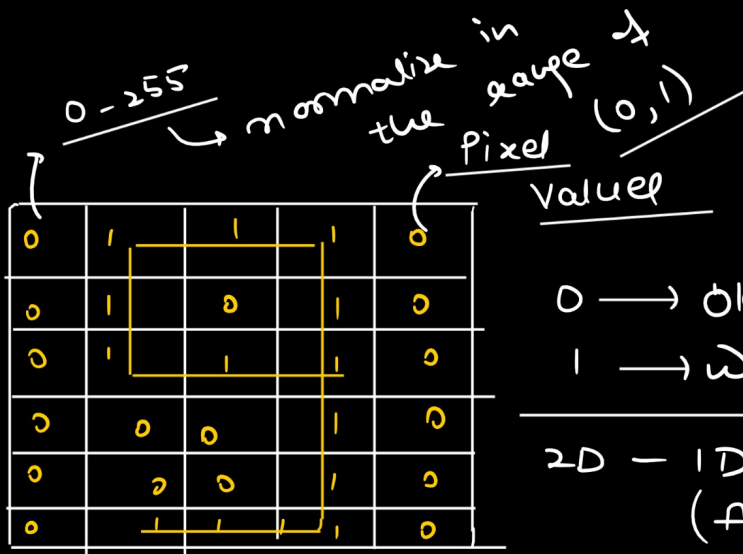
multiclassification task

↳ Softmax activation

function

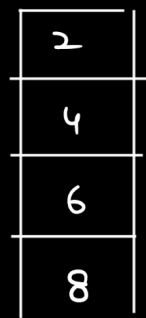
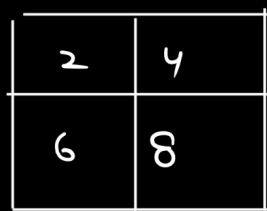
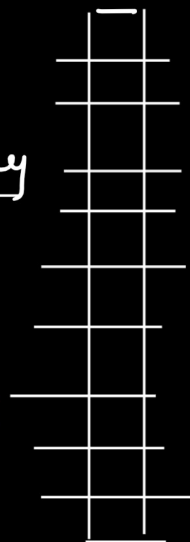
MNIST data → neural network

from scratch



Input layer

1D Array



→ Process these
values

inside the
neural network

Model: "sequential"

Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 784)	0
dense (Dense)	(None, 128)	100,480
dense_1 (Dense)	(None, 64)	8,256
dense_2 (Dense)	(None, 10)	650

Total params: 109,386 (427.29 KB)
 Trainable params: 109,386 (427.29 KB)
 Non-trainable params: 0 (0.00 B)

trainable

parameters

bias

weights $\rightarrow 64 * 10 + 10$ bias
 $\rightarrow 650$

weights $\rightarrow 128 * 64 + 64$
 $\rightarrow 8256$ bias $\Rightarrow 100480$

weights $\rightarrow 784 * 128 + 128$

Activation function

① sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (\text{binary})$$

(probability)

classification
task

$$\sigma'(x) = \sigma(x) (1 - \sigma(x))$$

$[0, 0.25]$

chain rule $\rightarrow$ differentiation

$$w_{\text{new}} = w_{\text{old}} -$$

$$\eta * \frac{\delta L}{\delta w_{\text{old}}}$$

vanishing gradient
problem

tanh activation function

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

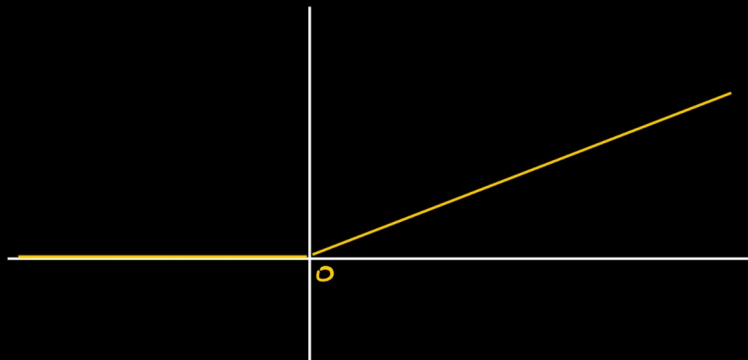
└──────────→ -1 to +1

$x \rightarrow \tanh(x)$

ReLU Activation function

$$f(x) = \max(0, x) \quad (\text{hidden layer})$$

└── faster computation



Softmax → multiple classes

└── probability

output layer

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum e^{z_j}}$$

$x \rightarrow -3, -1, 0, 2, 5$

mp. maximum

-3	_____	0	_____→	0
-1	_____	0	_____→	0
0	_____	0	_____→	0
2	_____	0	_____→	2
5	_____	0	_____→	5

element_wise